Alexander McIntosh

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Summary

AI Consultant & Systems Architect delivering **enterprise conversational analytics**, **production voice & agentic workflows**, and **interpretable machine learning** at scale.

Experience

Pioneer Management Consulting, Consultant, Artificial Intelligence — Minneapolis, MN
Advising enterprise leadership and delivering custom AI architectures, conversational analytics, and agentic automation to drive sustainable business adoption. — June 2026 – present, 5 months

- Architect custom AI systems delivering scalable predictive analytics and real-time anomaly detection for enterprise clients.
- Engineer production conversational analytics pipelines and generative domain agents.
- Embed model explainability and telemetry into system designs to ensure stakeholder trust and risk governance.

Optum, AI/ML Engineer Associate — Eden Prairie, MN
Architected and built enterprise-scale agentic voice AI systems with end-to-end observability, delivering AI tools to 400K+ employees. — Aug 2025 – June 2026, 11 months

- Shipped an agentic voice AI frontend using LiveKit to expose a conversational frontend to a multi-agent system
- Automated performance evaluation for LLM agents and RAG pipelines
- Shipped a multi-agent system using A2A, MCP, and ADK to automate admin tasks

Optum, Data Engineer Associate — Eden Prairie, MN
Architected and deployed enterprise-scale agentic AI systems with end-to-end observability, enhancing productivity for over 400K employees — Aug 2024 – Aug 2025, 1 year 1 month

- Automated LLM evaluation with scoring pipelines to monitor consistency, accuracy, safety, latency, and fairness
- Refactored Fabric Spark ETL into reusable patterns, cutting failure rates from 12-15% to <1%
- Mentored an intern team that won first place in the Optum-wide Shark Tank competition with an agentic AI solution

Optum, Data Science and Engineering Associate — Eden Prairie, MN
Developed and operationalized ML models and multi-agent systems to drive analytics and improve enterprise decision-making. — July 2023 – July 2024, 1 year 1 month

- Trained a state-of-the-art time-series forecasting model (Prophet + Random Forest) to detect operational anomalies with >90% precision.
- Orchestrated multi-agent AI systems to perform statistical analysis dashboard generation
- Delivered a conversational AI self-service analytics solution

Education

Master of Science — **Georgia Institute of Technology**, Computer Science — July 2027
Machine Learning specialization

MicroMasters — **MITx**, Statistics and Data Science — Dec 2023

Bachelor of Arts — **Saint John's University**, Physics — May 2021
Minored in Music & Mathematics • Phi Beta Kappa • Men's Chorus President

Production Infrastructure Mlops

Production Infrastructure & MLOps: Docker, OpenTelemetry, FastAPI, GitHub Actions (CI/CD), Microsoft Azure, Google Cloud Platform, Git

Skills

Datascience and ML: Python, SQL, R, PyTorch, TensorFlow, Scikit-learn, PySpark, MLFlow, Statistics and Probability

Agentic AI & RAG: Model Context Protocol (MCP), Agent Development Kit (ADK), Agent-to-Agent (A2A) Protocol, Semantic Kernel, Retrieval-Augmented Generation (RAG), LiveKit

Data & Cloud: Microsoft Fabric, Databricks, Delta Lake, AirFlow, ChromaDB, Git, Docker, Microsoft Azure, Google Cloud Platform, OpenTelemetry